Biraj Shrestha

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LinkedIn GitHub

Professional Summary:

Systems-focused Software Developer with hands-on experience in backend development, database optimization, and low-level systems programming. Strong foundation in C++ and PHP with practical experience building filesystem utilities, optimizing SQL performance, and deploying production backend services. Focused on building reliable, high-performance systems.

Education:

Bachelor of Science in Computer Science, **Youngstown State University** Graduated: December 2025

- GPA: 3.72/4.0, Dean's List, President's List
- Minor: Mathematics, Data Analytics Certification

Work Experience:

Junior PHP Developer, CloudRX, Framers Branch, TX (Feb 2026-Present)

- Worked within Git-based CI/CD deployment workflows across development, staging, and production environments.
- Resolved production tickets across backend and frontend systems using PHP, MySQL, and JavaScript, improving system reliability and reducing recurring data inconsistencies.
- Improved SQL query performance by restructuring date-range filters to preserve index usage and eliminating costly HAVING-based filtering, resulting in a ~15% reduction in execution time.
- Designed and implemented automated patient messaging workflows in PHP to schedule and dispatch notifications based on patient location and compliant communication windows (8AM–8PM). Integrated cron-based background processing services to reliably execute scheduled message delivery, reducing manual intervention and increasing system reliability.
- Investigated, debugged, and resolved production issues in a live healthcare environment through timely hotfix deployments, ensuring high system availability and uninterrupted business operations.

Student Programmer 1, Youngstown State University, Youngstown, OH (Fall 2024-Summer 2025)

- Developed and maintained custom PL/SQL applications integrated with Banner ERP to automate data retrieval for business operations across all university departments, reducing manual data lookup time from hours to minutes
- Collaborated directly with department stakeholders to gather requirements, test applications, and implement improvements through iterative feedback sessions
- Managed code versioning with TortoiseGit and deployed applications to production database via job submission workflow, coordinating with data custodians for release management
- Processed ticket-based requests using agile workflows, working with manager to prioritize and deliver solutions aligned with client needs

Technical Skills:

- Programming Language: C++ (11/14/17), Java, Python, PHP
- Backend: Python (API integration, data scraping, data pipelines, Fast API), Java Spring Boot (RESTful APIs, request handling, service-layer architecture)
- Database: SQL, MYSQL, SQLite, MSSQL
- Web Development: HTML, CSS, JavaScript
- Data Analysis: R, Tableau
- Tools: CMake, Make, Git, GitHub, Docker, TortoiseGit, PL/SQL Developer, Jira

Projects:

Virtual Disk File System Handler, (C++|Operating Systems|ext2)

- Engineered a C++ utility to perform high-speed file transfers between host systems and VirtualBox VDI disk images containing a Linux ext2 filesystem
- Deciphered complex disk layouts by parsing VDI headers and ext2 superblocks, navigating nested directory structures via inode traversal
- Implemented bitmap-based block allocation and manual direct/indirect block pointer management within ext2 inode structures
- Validated filesystem consistency by implementing metadata error-checking, preventing corruption during concurrent read/write operations.

Job Board Scrapper and Notifier, Python (Fast API, SQLite)

- Engineered automated job aggregation system integrating Adzuna(or any) Jobs API to fetch and process 200+ daily listings.
- Built production-grade RESTful API with FastAPI supporting filtered queries, pagination, and application tracking, achieving <10ms average response time and proper HTTP semantics (200, 404, 422 status codes)
- Optimized SQLite database performance through strategic B-tree indexing on high-cardinality columns (title, location, posted date), reducing query latency by 60% for searches across 1,000+ job postings
- Automated with cron scheduling (every 6 hours), enabling real-time Discord/email notifications and reducing manual job search time by 90%